



## OPEN The optimization of vocal music teaching by integrating the STEAM concept with the intelligent recommendation system

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To improve the personalization and intelligence level of vocal music teaching, this study integrates the Science, Technology, Engineering, Arts, and Mathematics (STEAM) concept into an intelligent recommendation system. It proposes a teaching optimization model based on multimodal learning and sentiment analysis (SA). The study comprehensively applies Neural Collaborative Filtering (NCF) to realize personalized recommendations, Deep Q-Network (DQN) to optimize teaching strategies, and Generative Adversarial Network (GAN) to generate diverse resources. It also combines multimodal fusion and SA to achieve real-time evaluation. The experiment is based on public data sources such as LibriSpeech, YouTube-8 M, Common Voice, and TED-LIUM. The results show that this model outperforms traditional methods in recommendation precision (F1-Score: 0.88), teaching strategy stability (97.24%), resource generation quality (97.91%), and multimodal fusion accuracy (99.79%). The study demonstrates the advantages of the in-depth integration of the STEAM concept and artificial intelligence. At the same time, it provides a practical new path for optimizing and promoting vocal music teaching. However, the real-time synchronization and deep semantic alignment among multimodal features still inevitably have certain limitations due to the computational complexity of existing algorithms and the limitations of model generalization abilities. In the future, a lightweight architecture and adaptive constraint mechanism can be combined to gradually improve the relevant technical paths.

**Keywords** Optimization of vocal music teaching, Neural collaborative filtering, Deep Q-Network, Generative adversarial network, STEAM concept

### Research background and motivations

In recent years, with the swift development of artificial intelligence (AI) and big data technology, the traditional vocal music teaching model has faced many challenges<sup>1</sup>. The demand for personalized and intelligent teaching is increasing, but existing systems still have shortcomings in processing multimodal data, real-time feedback, and sentiment analysis (SA)<sup>2,3</sup>. Therefore, this study integrates the Science, Technology, Engineering, Arts, and Mathematics (STEAM) concept into an intelligent recommendation system. The vocal music teaching process is optimized to enhance the teaching effect and student engagement by combining multimodal learning (ML) and SA methods<sup>4–6</sup>. An efficient, robust, and practical optimization model for vocal music teaching is implemented by integrating advanced algorithms and open data sources.

### Research objectives

Although existing studies on vocal music teaching optimization have achieved certain results in introducing personalized recommendations, ML, and AI methods, there are still obvious shortcomings. On the one hand, the in-depth fusion of multimodal data is insufficient. Multi-source information, such as audio, video, and text, is often used in isolation, and a unified feature modeling framework is lacking. On the other hand, current intelligent recommendation systems mostly stay at static matching or single-dimensional optimization, failing to dynamically capture and real-time adjust students' learning status and emotional feedback. In addition, existing models have limited performance in handling long-tail user behaviors and extreme data, and it is hard to balance personalization and robustness. This restricts the effect of teaching optimization in practical applications. These

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shortcomings indicate that the intelligent development of vocal music teaching urgently needs a systematic method that can balance multimodal data fusion, real-time feedback, and strategy optimization.

Grounded on this, the study constructs an optimization model of vocal music teaching driven by ML and SA by combining the STEAM concept with an intelligent recommendation system. Specific objectives include improving the accuracy of personalized recommendations, optimizing teaching strategies through Deep Q-Network (DQN), and using the Generative Adversarial Network (GAN) to enhance teaching resources' diversity and quality. In addition, multimodal data and emotional feedback are integrated to facilitate the teaching process's real-time response and guidance effect. Finally, a remarkable improvement in teaching effect is achieved, and student engagement is enhanced<sup>7</sup>. This design breaks through the limitations of traditional models in single modality, static recommendation, and limited feedback. It provides vocal music teaching with a complete integrated solution that can achieve "accurate matching - dynamic optimization - resource expansion - real-time evaluation". Thus, an overall improvement can be realized in teaching effectiveness, student engagement, and system intelligence levels.

## Literature review

### The application of AI and ML in music education

Over the years, the integration of AI and ML in education and music has gradually become a research hotspot. Chang et al. (2024) proposed the MusicARLtrans Net model, which integrated speech-to-text and Reinforcement Learning (RL) algorithms. This model achieved an accuracy of 96.77% on the LibriSpeech dataset and markedly improved the real-time feedback effect of interactive music education<sup>8</sup>. Sharif and Uckelmann (2024) introduced the Knowledge-driven and INtelligent Guidance for Holistic Teaching (KNIGHT) framework in education. They explored the application of multimodal data integration and Deep Reinforcement Learning (DRL) in educational analysis. Meanwhile, they focused on the issue of privacy-protected personalized feedback and revealed the complexity of educational technology in the digital context<sup>9</sup>. Govea et al. (2024) realized real-time emotion detection and personalized teaching strategies through DRL. The emotion recognition accuracy increased from 72.4% to 89.3%, and the personalization level rose from 70.2% to 90.1%. This indicated that emotion-driven changing adjustment played a key role in learning effectiveness<sup>10</sup>. Zaman et al. (2025) combined an Efficient Convolutional Neural Network (EfficientNet) with a convolutional neural network (CNN) optimized by transfer learning. This combination achieved an accuracy of 94.36% on the Facial Expression Recognition 2013 (FER-2013) dataset for human emotion recognition tasks, further strengthening the technical foundation of multimodal emotion feedback modeling<sup>11</sup>. At the same time, Zhang (2025) verified through comparative experiments that STEAM courses could effectively promote students' interdisciplinary knowledge transfer in music teaching. This enabled students to show stronger comprehensive abilities in lyric interpretation, understanding of creative backgrounds, and aesthetic appreciation, highlighting the STEAM concept's in-depth value for music education<sup>12</sup>. Besides, Daneshfar et al. (2024) systematically reviewed semi-supervised graph clustering technology, pointing out its potential in complex data analysis. They also provided methodological inspiration for the structured processing of multimodal data in education and art fields<sup>13</sup>.

Meanwhile, Keshun et al. (2024) showed new progress in the research of multimodal and physical information fusion. They proposed a fusion framework that combined acoustic vibration signals, physical constraints, and deep learning models. By embedding physical laws such as kinetic equations and energy conservation into the loss function, the model was guided to follow the constraint relationships of the real system during the feature learning process. Thus, pure data-driven models' overfitting and non-interpretability problems could be avoided<sup>14–17</sup>. This method realizes the dynamic correlation modeling between multi-modal signals while strengthening the expression of causal dependence between different signals through the attention mechanism. It enables the model to perform feature allocation and semantic interpretation under physical consistency. Although this direction mainly serves the state recognition of mechanical systems, its idea of "multi-modal representation guided by physical constraints" can also be transferred to the field of vocal music teaching. It can be used to construct a collaborative modeling mechanism for audio, video, and emotional feedback, thereby improving feature interpretability and the scientific nature of teaching feedback.

### Research progress of recommendation systems and generative models in teaching optimization

In terms of audio and video generation as well as educational resource expansion, Ahmad et al. (2025) proposed Deformable Periodic Network GAN (DPN-GAN). This model effectively improved the resolution and fidelity of audio synthesis by introducing periodic Rectified Linear Unit (ReLU) activation and deformable convolutional networks. It outperformed mainstream GAN architectures on multiple datasets, offering a reference for the generation of high-quality vocal practice resources<sup>18</sup>. Tong et al. (2024) proposed the Wav2Vid system, which used GAN to generate video avatars from audio. While maintaining perceived quality, it reduced the transmitted data volume by up to 83%, demonstrating the potential to reduce redundant data in distance education and interactive scenarios<sup>19</sup>. Bethencourt-Aguilar et al. (2023) verified the practical value of GAN in educational resource generation, with user satisfaction reaching as high as 92.1%<sup>20</sup>. In terms of recommendation and personalization, Prabhakar and Lee (2023) analyzed students' audio based on convolutional neural networks. It achieved a recommendation accuracy of 82.5% and promoted the development of behavior data-based learning recommendation systems<sup>21</sup>. The multimodal fusion model constructed by Chango et al. (2022) achieved a fusion accuracy of 94.2% after integrating audio, video, and text information. This illustrated the advantages of multimodal data collaboration in learning analysis<sup>22</sup>. Prottasha et al. (2022) used Bidirectional Encoder Representations from Transformers (BERT) to conduct SA, achieving a classification accuracy of 88.7% and improving the effectiveness of intelligent feedback systems<sup>23</sup>. The DQN optimization method proposed by Oroojlooyjadid et al. (2022) simulated the learning process, increasing the cumulative reward by 15.3%.

This verified the potential of RL in dynamic teaching strategy optimization<sup>24</sup>. Meanwhile, Vaziri et al. (2024) conducted a comprehensive evaluation of various machine learning methods. They emphasized the advantages and disadvantages of different models in multi-objective tasks and provided a reference for selecting multimodal algorithms in educational scenarios<sup>25</sup>. Overall, the above studies have gradually promoted the in-depth integration of music education and AI technology from multiple perspectives, including recommendation, generation, emotion recognition, and strategy optimization.

A conclusion can be drawn from existing studies. AI-driven personalized recommendation, multimodal fusion, and RL have achieved substantial progress in education and music teaching, and can significantly improve learning experience and teaching efficiency. This emphasizes the core role of cross-modal data integration, intelligent recommendation, and emotion recognition in future teaching optimization. However, most of the existing achievements focus on performance improvement of a single task or module, such as improving recommendation accuracy, enhancing audio synthesis quality, or optimizing emotion detection accuracy. They lack systematic integration of multimodal interaction, sentiment-driven feedback, and resource generation. At the same time, there are still distinct deficiencies in handling long-tail user behaviors, extreme data, and teaching robustness. Facing the growing demand for personalization and real-time performance, existing studies mostly stay at “local optimization” and fail to form a complete teaching optimization loop. In real vocal music teaching scenarios, however, it is often necessary to balance artistic expression and scientific optimization. Therefore, this study proposes an intelligent recommendation-driven vocal music teaching optimization model that integrates the STEAM concept. It uses Neural Collaborative Filtering (NCF) to improve the accuracy of personalized recommendations; DQN is utilized to realize optimization of strategies; GAN is employed to expand high-quality and diverse teaching resources. It also combines attention mechanism-based ML and SA to achieve real-time feedback, to build an integrated framework of “accurate matching - dynamic optimization - resource expansion - real-time evaluation”. This fills the gaps in existing studies and promotes the intelligent and personalized development of vocal music teaching.

## Research model

### The STEAM concept

The STEAM concept refers to integrating Science, Technology, Engineering, Arts, and Mathematics in education. Students’ innovative thinking and problem-solving ability are cultivated through the interdisciplinary integrated teaching mode. This concept emphasizes the integration of disciplines and uses art as a medium to stimulate students’ creativity and practical ability, thus promoting all-around development and knowledge application. The STEAM concept’s key features and representations are presented in Fig. 1.

### Related algorithms of the optimization model for vocal music teaching

#### The neural collaborative filtering (NCF) algorithm

The NCF algorithm is a recommendation algorithm that integrates neural networks and collaborative filtering technology. By capturing the non-linear interaction between users and projects, a Multilayer Perceptron (MLP) is used to model complex user preference patterns to improve the recommendation system’s accuracy and effect<sup>26</sup>.

The core calculation of personalized recommendation based on NCF is:

(1) Calculation of user embedding vector

$$P_u = W_P \cdot \left( \frac{x_u}{\|x_u\|} \right) + b_P \quad (1)$$

$P_u$  means the embedding vector of user  $u$ ;  $W_P$  refers to the user embedding matrix;  $\frac{x_u}{\|x_u\|}$  indicates the normalized user feature vector;  $b_P$  stands for the user embedding vector to calculate the bias vector.

(2) Computation of item embedding vectors

$$Q_i = W_Q \cdot \left( \frac{y_i}{\|y_i\|} \right) + b_Q \quad (2)$$

$Q_i$  refers to the embedding vector of item  $i$ ;  $W_Q$  represents the item embedding matrix;  $\frac{y_i}{\|y_i\|}$  is the normalized item eigenvector;  $b_Q$  means the item embedding vector to calculate the bias vector.

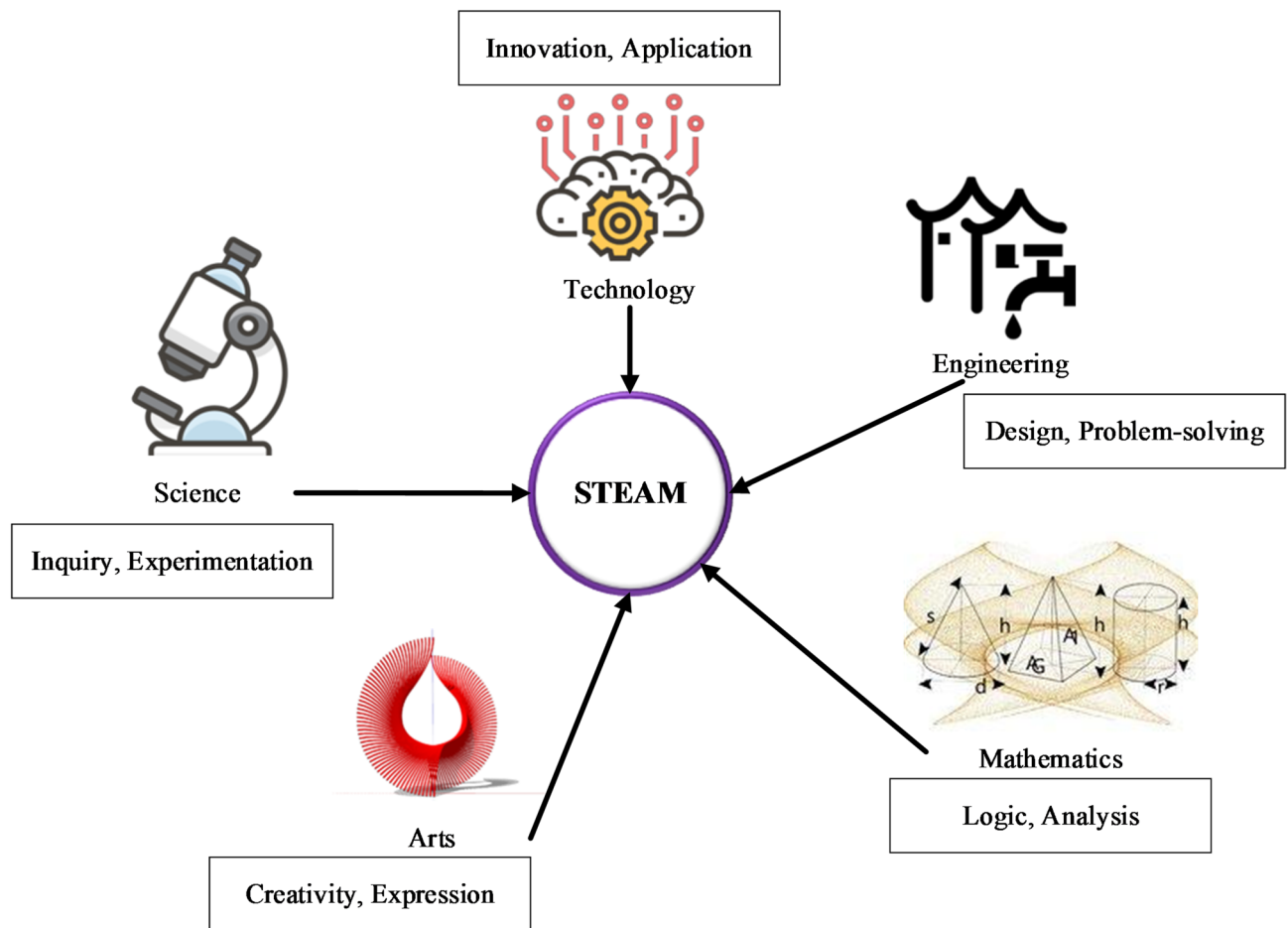
(3) Interactive computation of embedding vectors

$$z_{ui} = \text{concat}((P_u \circ Q_i), (P_u + Q_i), |P_u - Q_i|) \quad (3)$$

$z_{ui}$  represents the connected representation of the user and the project embedding vector;  $\circ$  denotes the Hadamard product (element product);  $|P_u - Q_i|$  means the absolute value of the element, and *concat* refers to the vector join operation.

(4) Computation of MLP

$$h_l = f \left( \frac{W_l \cdot h_{l-1} + b_l}{\sqrt{\sum_{k=1}^{d_{l-1}} (W_l \cdot h_{l-1})^2}} \right) \quad (4)$$



**Fig. 1.** The key features and representations of the STEAM concept.

$h_l$ ,  $W_l$ , and  $b_l$  represent the hidden vector, weight matrix, and bias vector of the  $l$ th layer;  $f$  is the activation function;  $h_{l-1}$  and  $d_{l-1}$  refer to the output and dimension of the previous layer.

(5) Calculation of the final prediction score

$$\hat{y}_{ui} = \sigma \left( \frac{h_L \cdot w_o + b_o}{1 + \exp(-(h_L \cdot w_o + b_o))} \right) \quad (5)$$

$\hat{y}_{ui}$  refers to the predicted score of the user  $u$  on item  $i$ ;  $h_L$  denotes the output of the last layer;  $w_o$  and  $b_o$  are the weight and bias of the output layer;  $\sigma$  is the sigmoid function.

In NCF, the calculation of user vector  $P_u$  and item vector  $Q_i$  is essentially a vectorized mapping of learner features and teaching resource features. Through normalization, the model can emphasize relative relationships rather than absolute differences in high-dimensional space. This allows students' learning behaviors and music piece features to be compared on a unified scale. This design does not simply pursue algorithm performance, but aligns with the scientific and mathematical dimensions in the STEAM framework: it transforms artistic performances into quantifiable structures, enabling individual differences to be modeled and explained. In the interaction representation of Eq. (3), the product, sum, and difference of vectors are considered simultaneously, reflecting the multi-level alignment and differences between students and music pieces. The modeling logic here has an inherent similarity to "style matching" in art education: instead of relying on a single indicator, it uses multi-angle mathematical operations to reveal potential learning paths. Finally, the MLP realizes non-linear mapping through Eqs. (4) and (5), further abstracting this complex relationship to form a recommendation probability. Its value lies in providing a mathematically driven interpretation framework for the teaching process, allowing art education to go beyond empirical judgments.

#### The DQN algorithm

The DQN algorithm combines the Deep Neural Network (DNN) and reinforcement learning. This algorithm also uses DNN to approximate the Q function (the action-value function) and learns the optimal strategy through environmental interaction<sup>27,28</sup>. Meanwhile, by simulating the interaction between students and teaching content, teaching strategies can be dynamically adjusted to achieve personalized teaching and improve learning efficiency<sup>29,30</sup>.

The core calculation of teaching strategy optimization by the DQN is as follows:

(1) Update of the Q value

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left( r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right) \quad (6)$$

$Q(s_t, a_t)$  represents the Q value of the  $a_t$  taken in the state  $s_t$ ;  $\alpha$  means the learning rate;  $r_t$  refers to the immediate reward;  $\gamma$  is the discount factor;  $\max_{a'} Q(s_{t+1}, a')$  denotes the maximum Q value of the next state  $s_{t+1}$ <sup>31</sup>.

(2) Loss function

$$\mathcal{L}(\theta) = \mathbb{E}_{(s, a, r, s')} \left[ \left( r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right] \quad (7)$$

$\mathcal{L}(\theta)$  means the loss function;  $\theta$  is the parameter of the Q network;  $\theta^-$  denotes the parameter of the target Q network;  $(s, a, r, s')$  represents the sample in the empirical replay<sup>32</sup>.

(3) Calculation of the target Q value

$$y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-) \quad (8)$$

$y_t$  stands for the target Q value.

(4) Update of experience replay pool

$$D \leftarrow D \cup \{(s_t, a_t, r_t, s_{t+1})\} \quad (9)$$

$D$  represents the experience replay pool;  $(s_t, a_t, r_t, s_{t+1})$  denotes the current experience.

(5) Parameter update

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta) \quad (10)$$

$\eta$  refers to the learning rate, and  $\nabla_{\theta} \mathcal{L}(\theta)$  is the gradient of the loss function.

(6) Expansion of the Bellman Eq.

$$Q(s_t, a_t) = \mathbb{E} \left[ r_t + \gamma \max_{a'} Q(s_{t+1}, a') | s_t, a_t \right] \quad (11)$$

In the core update Eq. (6) of DQN, the learning of state-action value depends on the quantitative description of students' performance, such as intonation deviation, rhythm stability, and timbre consistency. The introduction of these indicators is not only for strategy optimization, but also reflects the reconstruction of artistic subjectivity by scientific and engineering methods: the "good" and "bad" that originally relied on teachers' intuition are now translated into measurable signals. The loss function in Eq. (7) and the calculation of the target Q-value in Eq. (8) aim to continuously adjust the parameters of the strategy network, enabling the teaching process to dynamically follow changes in students' states. This mechanism actually corresponds to the "engineering-feedback" link in STEAM, that is, maintaining the system's stability and personalization through continuous testing and correction.

#### The GAN algorithm

GAN consists of generators and discriminators that generate high-quality vocal practice content through adversarial training. The generator is responsible for generating new practice resources; the discriminator is used to evaluate the authenticity and validity of the generated content<sup>33-35</sup>. The core calculation of resource generation based on GAN is:

(1) The loss function of the generator.

$$\mathcal{L}_G = -\mathbb{E}_{z \sim p_z(z)} [\log D(G(z))] \quad (12)$$

$\mathcal{L}_G$  stands for the loss function of the generator;  $G(z)$  represents the sample generated by the generator;  $D(G(z))$  means the discriminator's discrimination result on the generated sample;  $p_z(z)$  denotes the distribution of latent variables<sup>36</sup>.

(2) The loss function of the discriminator.

$$\mathcal{L}_D = -\mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] - \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))] \quad (13)$$

$\mathcal{L}_D$  means the loss function of the discriminator,  $x$  denotes the real sample, and  $p_{\text{data}}(x)$  represents the distribution of the real data.

(3) The weight update of the discriminator.

$$\theta_D \leftarrow \theta_D - \eta_D \nabla_{\theta_D} \mathcal{L}_D \quad (14)$$

$\theta_D$  and  $\eta_D$  refer to the weight parameter and learning rate of the discriminator;  $\nabla_{\theta_D} \mathcal{L}_D$  represents the gradient of the discriminator loss function to the weight parameter<sup>37–39</sup>.

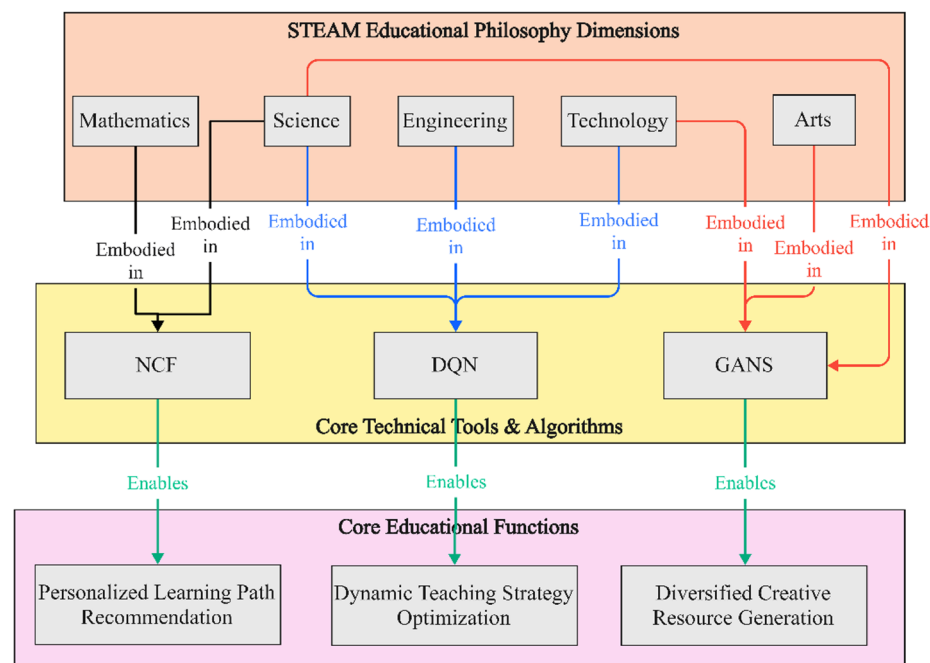
(4) Wasserstein GAN loss function

$$\mathcal{L}_D = \mathbb{E}_{x \sim p_{\text{data}}(x)} [D(x)] - \mathbb{E}_{z \sim p_z(z)} [D(G(z))] \quad (15)$$

In GAN, the adversarial training relationship between the generator and discriminator loss in Equations (12) Eq. (13) is not just an algorithmic game. Conversely, they can be understood as the tension between artistic creation and scientific verification. The generator tries to synthesize materials similar to real singing, while the discriminator provides an “objectified critique,” similar to the professional evaluation of a teacher. Through this dynamic balance, the system can form a continuously evolving resource library. The audio generated by GAN is used as practice materials and further input into the NCF recommendation module as candidate resources. The generator expands the boundaries of artistic resources, and the recommendation system integrates them into the overall path of personalized learning. Thus, scientific modeling, technical implementation, and artistic creation form a closed loop.

The mapping relationship among the STEAM concept, NCF/DQN/GAN technical tools, and educational functions is displayed in Fig. 2:

In Fig. 2, the STEAM concept is systematically transformed into the internal logical framework for algorithm design and teaching optimization in the proposed intelligent recommendation system. The Science dimension is reflected in the quantitative modeling and law extraction of the vocal music learning process. By establishing testable hypotheses and data feedback mechanisms, it converts teaching behaviors into an observable and verifiable scientific process. The Technology dimension is integrated into the multi-modal data fusion and knowledge representation system. It emphasizes the dynamic transmission and structural coordination of information flow, enabling the system to form stable semantic consistency and adaptive correlation at the auditory, visual, and textual levels. The Engineering dimension is embodied in the real-time feedback and continuous optimization mechanism of the system architecture. Through parameter constraints and self-adjustment of the objective function, it simulates strategy improvement and path optimization in teaching, and constructs a learning closed loop with self-evolution ability. The Arts dimension is not an additional rhetorical element in the design, but one of the core driving forces. The system incorporates emotional expression, aesthetic consistency, and expressive intention into the selection and generation of learning resources, making the recommendation results conform to logical efficiency and have artistic appeal and educational resonance. The Mathematics dimension runs through the system’s underlying structure and serves as the formal basis for realizing the above concepts. Whether in constructing embedded space, allocating feature weights, or verifying convergence and optimization solutions, mathematical logic provides the model with rigorous theoretical constraints and interpretability support. Thus, the STEAM concept no longer stays at the level of educational philosophy. Instead, through structural mapping



**Fig. 2.** The mapping relationship among the STEAM concept, NCF/DQN/GAN technical tools, and educational functions.



and function embedding, it becomes a generative framework and the core of innovative logic for the design of the recommendation system.

### The integrated system architecture of the vocal music teaching optimization model

Under the guidance of the STEAM concept, the constructed vocal music teaching optimization model is not a simple stack of single technologies. Instead, it forms a functional system centered on data-driven logic through the integration of scientific methods, engineering modeling, and art education. Specifically, the system first inputs audio, video, and text into the multimodal evaluation (ML/SA) module. Through feature extraction, attention mechanism, and SA, it realizes unified modeling of cross-modal features, thereby converting students' singing performance, emotional fluctuations, and learning status into quantifiable indicators. This objective data serves as real-time feedback to the teaching process and provides basic support for subsequent modules. Subsequently, the NCF personalized recommendation module uses the NCF method to achieve in-depth matching between students' learning features and teaching resource features. This ensures that the recommendation results can match students' skill levels and meet the needs of improving their artistic expression. At the same time, the GAN resource generation module expands diverse vocal practice materials through adversarial training. This alleviates the problems of limited traditional teaching resources and a single style, and continuously supplements the generated resources to the recommendation and evaluation links, enhancing the richness and creativity of learning. Regarding strategy optimization, the DQN module dynamically adjusts teaching strategies based on the learning status and emotional feedback from the teaching platform. This ensures that the teaching progress is compatible with students' emotional states, avoiding excessive frustration and preventing learning stagnation. Through the interactive flow of the above modules, the system realizes a closed-loop process from multimodal data collection, accurate recommendation, resource expansion to strategy optimization. It combines the interdisciplinary integration in the STEAM concept with the creative needs of art education, thereby solving the problem of over-reliance on teachers' experience and subjective judgment in traditional vocal music teaching. This enables artistic evaluation to be based on scientific data and dynamic optimization, ultimately promoting vocal music teaching to achieve objective, personalized, and intelligent development.

The integration implementation process of the proposed vocal music teaching optimization model's system architecture is shown in Fig. 3:

### Experimental design and performance evaluation

This study integrates the STEAM concept into the intelligent recommendation system to optimize vocal music teaching. It selects key algorithms from four major optimization directions: personalized recommendation, teaching strategy optimization, resource generation, and comprehensive assessment; it aims to enhance teaching effectiveness and student engagement. The specific optimization processes and implementation principles are as follows:

First, personalized recommendation utilizes DNN to process user data (learning progress and feedback) and item data (practice repertoire). Meanwhile, it constructs an interaction matrix and generates high-dimensional embeddings for users and items. Through MLP, it learns users' latent ratings for items and makes personalized teaching content recommendations based on predicted ratings, achieving precise matching.

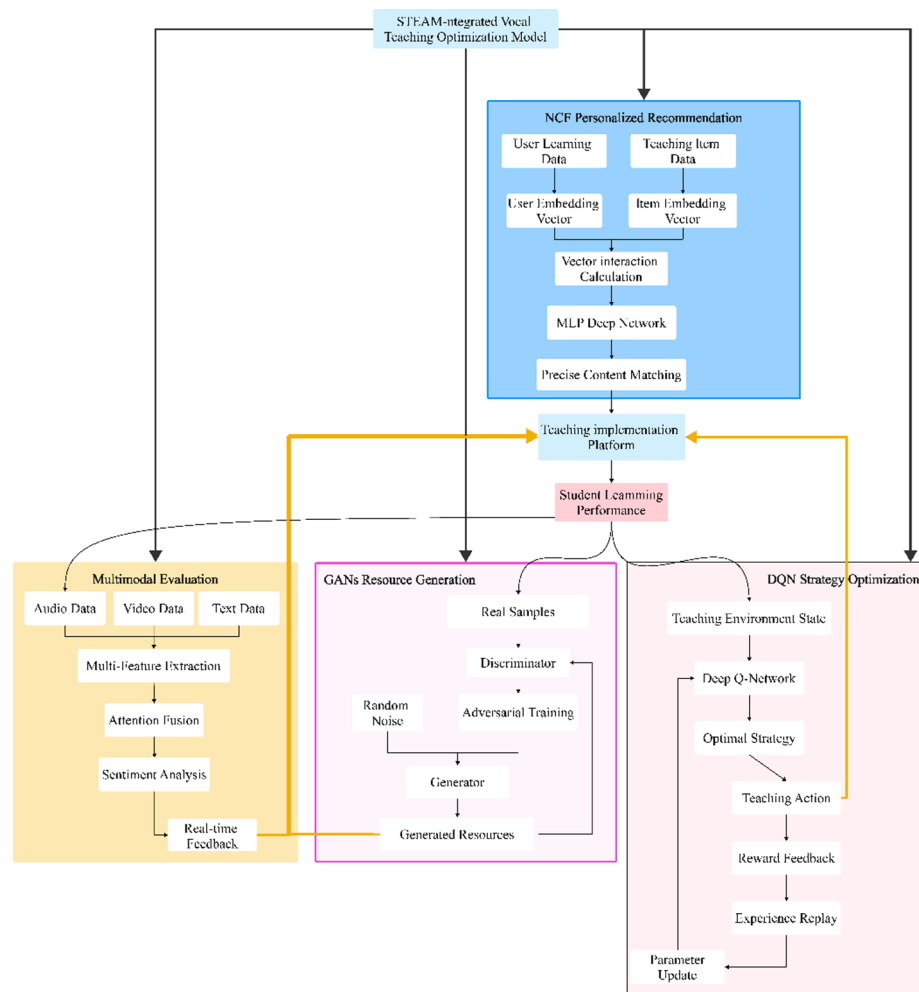
Second, modeling the teaching environment as a Markov Decision Process (MDP), the study employs the DQN algorithm to optimize teaching strategies for improving student learning outcomes and engagement. DQN continuously updates the state-action values approximated by the Q-function, utilizing techniques such as experience replay and target networks to optimize teaching strategies<sup>40–42</sup>. This process adjusts teaching content and methods in real-time to maximize student learning progress and satisfaction.

Regarding resource generation in vocal music teaching, the generator takes random noise as input to simulate vocal exercise audio. The discriminator provides feedback by comparing real and generated audio signals. Through iterative adversarial training, the generator improves its capability to produce increasingly realistic exercise resources, enriching teaching materials and enhancing student interest and engagement<sup>43,44</sup>.

In a comprehensive assessment, ML and SA are integrated to analyze audio, video, and text data, extracting features including facial expressions and actions. Attention mechanisms are used to fuse multimodal features and the BERT model for sentiment classification. Then, an evaluation model is constructed to offer real-time feedback on student learning outcomes and emotional states, guiding teaching optimizations.

### Datasets collection

During the model validation process, data collection and processing are conducted through public channels to ensure the avoidance of ethical issues. A total of no less than 300,000 pieces of audio, video, and text data are collected using large-scale open-source datasets such as LibriSpeech, YouTube-8 M, Common Voice, AVSpeech, TED-LIUM, and VoxCeleb. Compared with other open-source datasets that can be accurately adapted to vocal music types, the TED-LIUM and VoxCeleb general speech datasets are selected. This is because they cover a wide range of speaker characteristics and high-quality speech samples. These data can provide the model with diverse speech patterns and emotional features. Although they are not specialized vocal music datasets, their speech diversity and transferability help improve the model's generalization ability in vocal music teaching scenarios. Data preprocessing encompasses audio feature extraction, video frame segmentation, and others. Through standardization, denoising, and data enhancement techniques, data quality and diversity can support the efficiency and reliability of model training and validation<sup>45</sup>. In the data preprocessing stage, all multimodal data undergo a unified process of feature extraction and standardization. The LibriSpeech and Common Voice datasets are the main audio and text sources. They undergo silent segment truncation, amplitude normalization, and feature extraction (Mel-Frequency Cepstral Coefficients (MFCC), Chroma, Spectrogram), respectively. Text samples are processed through cleaning, lowercasing, and synonym replacement. Meanwhile, sample diversity



**Fig. 3.** The integration implementation process of the system architecture for the vocal music teaching optimization model.

is enhanced through standardization and corresponding augmentation strategies (such as noise addition, frequency masking, and semantic replacement). The YouTube-8 M and AVSpeech datasets mainly provide video and multimodal samples. They first undergo frame extraction, facial key point detection, and frame standardization. Then, CNN and 3D-CNN extract frame-level features, followed by brightness normalization and frame augmentation. The TED-LIUM dataset mainly consists of high-quality transcribed text. After text cleaning, lowercasing, tokenization, and synonym replacement, it is stratified by speaker ID to maintain the diversity and independence of the corpus. Before feature extraction, the VoxCeleb audio data uses Voice Activity Detection (VAD) and energy normalization to filter out segments with low signal-to-noise ratio. Subsequently, MFCC and Chroma features are extracted, and noise addition is adopted for data augmentation.

To ensure sample balance and representativeness, all datasets are divided into training set, validation set, and test set at a ratio of 8:1:1. For cross-data source tasks, five-fold cross-validation is used to reduce data bias. Input data in the training stage are all standardized by Z-score. The length of audio samples is unified to a 4-second window, the video frame sampling rate is set to 24 fps, and the text sequence length is truncated to 256 tokens. These measures ensure the dimensional consistency of multimodal features. After preprocessing and division, the total amount of data reaches 327,837 entries. The dataset information is exhibited in Table 1:

In the multi-source data integration stage, the six datasets are not used in isolation. Instead, they are mapped to unified learning units by aligning labels and timestamps. Specifically, audio samples correspond to vocal music segments, video samples correspond to performance movements or classroom clips, and text samples are associated with lyrics, teaching prompts, or student feedback. This multimodal information is uniformly labeled as teaching resource units, which form the “items” in the interaction matrix. Correspondingly, “users” refer to individual students. Their historical learning trajectories, singing records, and emotional expressions are mapped to the same space. This method abstracts data of different modalities into comparable units, allowing NCF to directly model the relationship between students and cross-modal resources.

In the unified modeling process, audio features (MFCC, Chroma, spectrogram), video frame features (expressions and movements extracted by CNN/3D-CNN), and text features (BERT-based embedding vectors) are aligned with the learning trajectories of individual students; it aims to construct a cross-modal “user-item”



| Data source  | Data type | Data volume | Preprocessing steps | Feature extraction methods | Standardization       | Data enhancement      |                   |
|--------------|-----------|-------------|---------------------|----------------------------|-----------------------|-----------------------|-------------------|
| LibriSpeech  | Audio     | 84,893      | Feature extraction  | MFCC, Chroma               | Standardization       | Noise addition        |                   |
| LibriSpeech  | Text      | 23,505      | Text cleaning       | Tokenization               | Lowercase             | Synonym replacement   |                   |
| YouTube-8 M  |           | Video       | 51,975              | Frame extraction           | CNN for Frames        | Frame standardization | Frame enhancement |
| YouTube-8 M  | Text      | 17,325      | Text vectorization  | BERT Embeddings            | /                     | /                     |                   |
| Common Voice | Audio     | 34,650      | Feature extraction  | Spectrogram                | Standardization       | Frequency mask        |                   |
| AVSpeech     | Video     | 41,580      | Frame extraction    | 3D-CNN                     | Frame standardization | Frame enhancement     |                   |
| TED-LIUM     | Text      | 27,692      | Text cleaning       | Tokenization               | Lowercase             | Synonym replacement   |                   |
| VoxCeleb     | Audio     | 46,217      | Feature extraction  | MFCC, Chroma               | Standardization       | Noise addition        |                   |

Table 1. The proposed dataset.

| Category             | Parameter                      | Specification         |
|----------------------|--------------------------------|-----------------------|
| Hardware environment | Central Processing Unit (CPU)  | Intel Xeon E5-2680 v4 |
|                      | Graphics Processing Unit (GPU) | NVIDIA Tesla V100     |
|                      | Memory                         | 128GB DDR4            |
| Software environment | Operating system               | Ubuntu 20.04 LTS      |
|                      | Programming language           | Python 3.8            |

Table 2. Experimental environment.

interaction matrix. Among them, “users” are defined as individual students, and “items” are specified as vocal music teaching resources. These resources include existing music pieces, practice segments, and teaching videos; meanwhile, extended materials are generated by GAN. After this processing, NCF can capture the interaction relationship between students and multimodal resources in a unified space. In the experimental implementation of DQN, the state  $s_t$  is mapped to the quantifiable performance of a student at a certain moment. This includes intonation deviation, rhythm stability, timbre consistency, and emotional tone (obtained through the joint analysis of the emotional color of text feedback and singing audio by BERT). The action  $a_t$  corresponds to the specific intervention method selected by the system in the next teaching step, such as recommending new practice pieces, adjusting difficulty, or providing emotional feedback. The reward  $r_t$  takes the comprehensive indicators of the students’ immediate performance improvement (e.g., reduced intonation error, improved rhythm stability) and subjective feedback scores. In the GAN part, the generator adopts a Conv-Deconv structure, which takes random noise vectors and simplified music score segments as input to generate vocal practice audio. The discriminator is based on a convolutional discriminant network, which distinguishes between generated samples and real audio, and uses feedback signals to optimize the generator in a reverse manner. Through this design, the system generates materials similar in style to the target music library and produces audio with differences in pitch, rhythm, and emotional color. This provides diverse resource support for subsequent recommendations and training links.

Experimental environment

The experimental environment is listed in Table 2:

Parameters setting

To further verify the contribution of each module to the overall system performance, a systematic ablation experiment is designed. The specific method involves removing or replacing key components one by one. (1) NCF is replaced with traditional CF to observe changes in the effect of personalized recommendation; (2) DQN is removed, leaving only static recommendations without dynamic teaching strategy optimization; (3) GAN is removed, relying only on real data resources; (4) ML/SA is removed, using only single-modal speech features as evaluation input. All experimental groups and the complete model run under the same environment and data conditions. The following indicators are calculated respectively. Recommendation accuracy: The F1-score is calculated based on the consistency between predicted results and actual interaction records. Strategy stability: It is measured by the fluctuation range of the cumulative reward curve in multiple training rounds. Generation diversity: The gap between generated and real samples is quantified based on the distribution difference of Mel spectrum features. Sentiment recognition accuracy: Determined by the consistency between the model’s classification results and the labels of public benchmark data. The results are listed in Table 3:

It can be found from the results that each module makes a remarkable contribution to the system performance. When NCF is replaced with traditional CF, the recommendation accuracy decreases most obviously (F1-score drops from 0.88 to 0.74), which indicates the necessity of deep modeling. Removing DQN leads to a significant decline in teaching strategy stability (dropping from 97.24% to 84.56%), demonstrating the key role of RL in real-time teaching feedback. After removing GAN, the resource generation diversity index drops to 0 directly, verifying the unique contribution of this module in expanding practice resources. Removing the ML/SA module causes a sharp decline in sentiment recognition accuracy (dropping from 99.79% to 83.64%), showing that single-

| Model settings      | Recommendation accuracy | Strategy stability (%) | Generation diversity index | Sentiment recognition accuracy (%) |
|---------------------|-------------------------|------------------------|----------------------------|------------------------------------|
| Complete model      | 0.88                    | 97.24                  | 0.93                       | 99.79                              |
| Replace NCF with CF | 0.74                    | 96.81                  | 0.91                       | 99.65                              |
| Remove DQN          | 0.87                    | 84.56                  | 0.92                       | 99.72                              |
| Remove GAN          | 0.85                    | 96.93                  | 0.00                       | 99.77                              |
| Remove ML/SA        | 0.87                    | 97.15                  | 0.92                       | 83.64                              |

**Table 3.** Comparison of ablation experiment results.

| Parameter             | Value                       | Description                                                                  |
|-----------------------|-----------------------------|------------------------------------------------------------------------------|
| Training epochs       | 100 epochs                  | The length of the model training epochs, used to observe long-term stability |
| Batch size changes    | ± 32                        | The impact of batch size changes on training effectiveness                   |
| Data noise levels     | ± 5%                        | Changes in noise levels introduced in input data                             |
| Model convergence     | Convergence speed and error | Convergence speed and error analysis of models under different conditions    |
| Parameter sensitivity | Performance fluctuations    | Sensitivity of the model to parameter changes and performance fluctuations   |

**Table 4.** Robustness test parameters.

| Parameter              | Value        |
|------------------------|--------------|
| User embedded size     | 32           |
| Project embedding size | 32           |
| Hidden layer           | [64, 32, 16] |
| Activation function    | ReLU         |
| Optimizer              | Adam         |
| Training data volume   | 200,000      |

**Table 5.** The NCF parameters of the proposed model in the personalized recommendation validation.

modal evaluation is difficult to capture complex sentiment features. Overall, the four modules complement each other and jointly ensure the integrity and advantages of the system in personalized recommendation, dynamic optimization, resource expansion, and sentiment feedback.

During the formal verification process, the optimized system’s stability is first analyzed by examining parameters such as training epochs, learning rate variations, batch size changes, and data noise levels. Subsequently, model convergence and parameter sensitivity are assessed to evaluate the system’s stability and robustness under diverse conditions. This ensures that the model maintains high performance and consistency across various scenarios. The parameters for robustness testing are given in Table 4:

Second, the study uses six publicly available data sources, including LibriSpeech, YouTube-8 M, Common Voice, AVSpeech, TED-LIUM, and VoxCeleb. The effectiveness of personalized recommendations is validated across six groups of data. Each group covers multimodal information such as student learning progress, audio features, and textual feedback. Personalized recommendations are made by constructing user-item interaction matrices and employing the NCF algorithm to generate a high-dimensional embedded representation. The optimization process is assessed using indicators such as Mean Absolute Error (MAE), precision, recall, F1-score, and Root Mean Square Error (RMSE) to demonstrate the advantages of personalized recommendations in vocal music teaching. Among them, the F1 value is calculated based on the harmonic mean of Precision and Recall; MAE and RMSE measure the error between the predicted score and the actual feedback, respectively. The used NCF parameters are detailed in Table 5.

Third, the effectiveness of DQN in optimizing teaching strategies is evaluated based on cumulative rewards, learning curves, and strategy stability indicators. By simulating learning processes and adjusting parameters, the convergence and optimization effects of DQN are verified to ensure the effectiveness and stability of optimized strategies. To ensure the interpretability and stability of the optimization of teaching strategies, this study designs a composite reward function in the DQN.

$$R_t = \alpha A_t + \beta P_t - \lambda L_t \tag{16}$$

$A_t$  represents the learner’s immediate performance improvement in the current round (such as the reduction rate of intonation error and the improvement rate of rhythm stability);  $P_t$  denotes the teaching path matching degree (calculated from the similarity between the recommendation result and the learning goal);  $L_t$  is a strategy fluctuation penalty term, which is used to suppress excessive exploration. The parameter weights are set to  $\alpha=0.4$ ,  $\beta=0.5$ , and  $\lambda=0.1$  to balance individual performance and strategy stability. In the result analysis, independent sample t-tests and one-way Analysis of Variance (ANOVA) are used to verify the statistical

| Parameter               | Value   |
|-------------------------|---------|
| State size              | 128     |
| Action size             | 10      |
| Playback buffer size    | 15,000  |
| Discount factor (Gamma) | 0.95    |
| Target update frequency | 2000    |
| Maximum exploration     | 1       |
| Minimum exploration     | 0.01    |
| Explore attenuation     | 0.99    |
| Training set size       | 150,000 |
| Verification set size   | 62,837  |

**Table 6.** The proposed model’s DQN parameters in teaching strategy optimization verification.

| Parameter                  | Value            |
|----------------------------|------------------|
| Generator input size       | 128              |
| Generator hidden layer     | [256, 512, 1024] |
| Discriminator hidden layer | [1024, 512]      |
| Activation function        | LeakyReLU        |
| Optimizer                  | RMSprop          |

**Table 7.** GAN parameters of the proposed model in resource generation effect validation.

| Names of indicators                   | Explanation of meaning                                                                                              |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Audio feature extraction accuracy (%) | The accuracy of acoustic feature recognition by the system during the audio processing phase                        |
| Video analysis accuracy (%)           | The accuracy of expression and action recognition by the system in video frames                                     |
| Multimodal fusion accuracy (%)        | The system’s overall discriminative accuracy after integrating audio, video, and text features                      |
| Sentiment classification accuracy (%) | The accuracy of the system in recognizing students’ emotional states                                                |
| Real-time feedback accuracy (%)       | The system’s consistency in providing immediate teaching feedback with standard results                             |
| Teaching guidance improvement (%)     | The improvement rate in learning progress and teaching effectiveness after system optimization                      |
| Overall system efficiency (%)         | The system’s comprehensive efficiency in completing recommendations, generation, and feedback within a unit of time |

**Table 8.** An explanation of the relevant indicators for comparing the comprehensive evaluation effects of the proposed model and baseline models.

significance of the DQN optimization effect. The two methods conduct significance tests on cumulative rewards, learning curve slopes, and strategy convergence values, with the significance level set to  $p<0.05$ . Statistical results show that the optimized DQN model is remarkably superior to the baseline model in convergence speed and strategy stability; this verifies the effectiveness and robustness of the teaching strategy optimization. Other key parameters of DQN in the study are detailed in Table 6:

Fourth, the GAN’s performance in generating high-quality and diverse vocal exercise resources is evaluated using indicators such as generation quality, diversity, and user satisfaction. Combining signal processing techniques and subjective evaluations, the study analyzes the practical value of GAN-generated resources in educational practices. The used GAN parameters are drawn in Table 7.

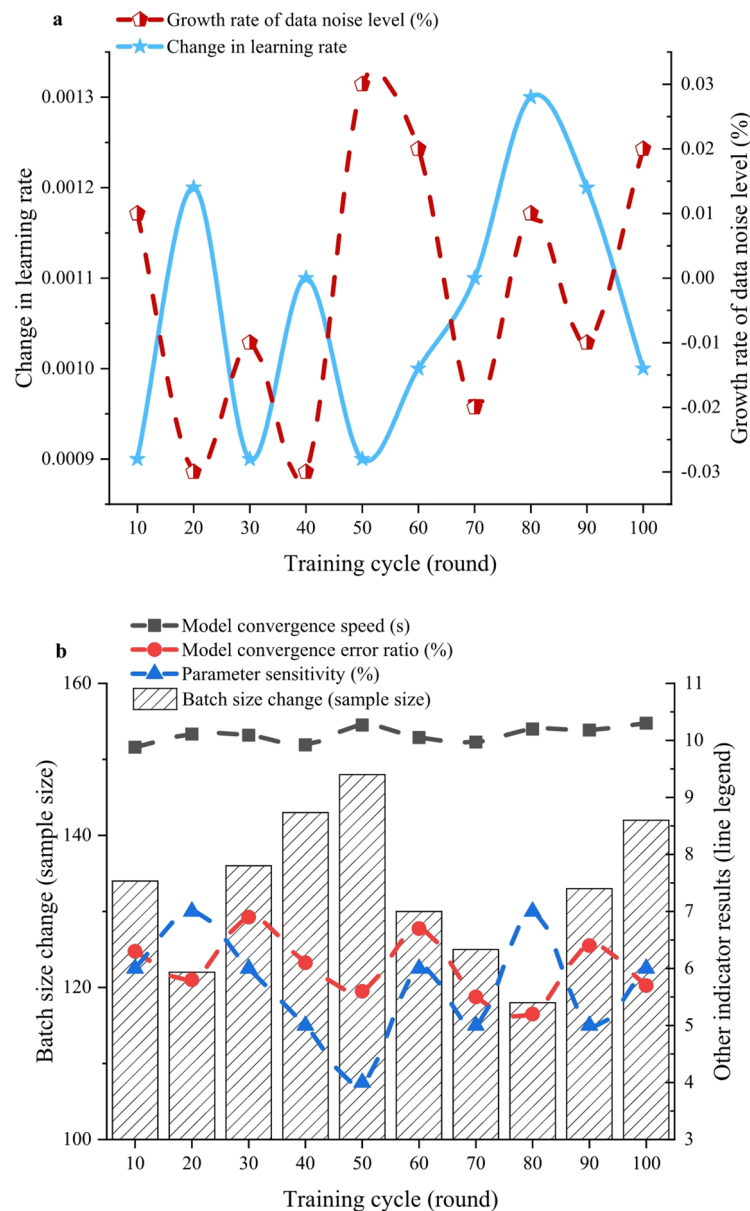
Fifth, through indicators such as audio feature extraction accuracy and video analysis accuracy, the comprehensive evaluation effect of the proposed model and baseline models is compared. The explanation of the indicators selected in the study is depicted in Table 8.

Performance evaluation

Robustness test analysis

The robustness analysis results of the complete training epoch-based vocal music teaching optimization model are presented in Fig. 4:

Figure 4 denotes that within 100 training epochs, the learning rate varies between 0.0009 and 0.0013, batch sizes range from 118 to 148 samples, and data noise level changes are between  $-0.03\%$  and  $0.03\%$ . Moreover, model convergence speeds are between 9.88 s and 10.3 s, convergence errors are between 5.2% and 6.9%, and parameter sensitivity ranges from 4% to 7%.



**Fig. 4.** Robustness analysis of the vocal music teaching optimization model by the complete training epoch ((a): Results of changes in learning rate and data noise level; (b): Results of model convergence and sensitivity).

#### The personalized recommendation effect analysis

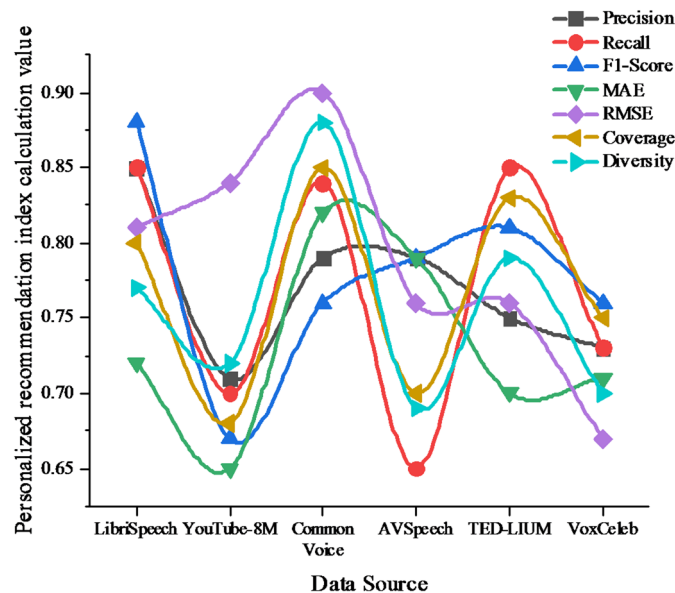
The analysis results of the personalized recommendation effect of the vocal music teaching optimization model based on six datasets are revealed in Fig. 5:

In Fig. 5, based on the LibriSpeech data source, the recommendation model achieves outstanding performance with Precision and Recall reaching 0.85 each. The Common Voice data source excels in F1-Score and Coverage, at 0.76 and 0.85, respectively. The TED-LIUM data source performs well in MAE and Diversity, scoring 0.70 and 0.79.

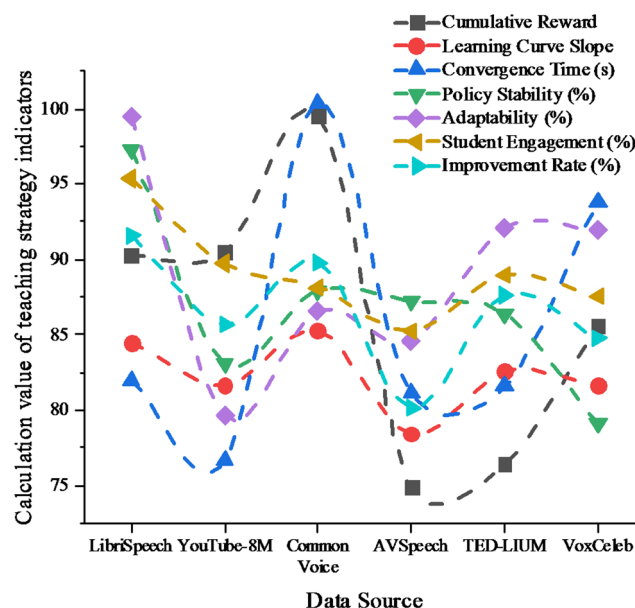
#### Analysis of teaching strategy optimization effect

The analysis results of the teaching strategy optimization effect of the six datasets-based vocal music teaching optimization model are illustrated in Fig. 6:

Figure 6 displays that, based on the LibriSpeech data source, the recommendation model exhibits excellent performance with Policy Stability at 97.24% and Adaptability at 99.46%. The Common Voice data source performs exceptionally well with Cumulative Reward at 99.51% and Convergence Time at 100.41 s. The TED-LIUM data source shows strong performance in Student Engagement at 88.97% and Improvement Rate at 87.63%.



**Fig. 5.** The personalized recommendation effect analysis of the vocal music teaching optimization model based on six datasets.



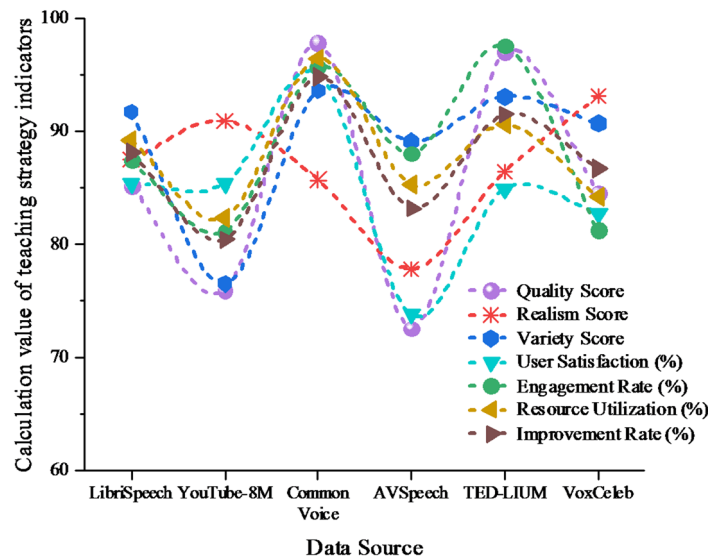
**Fig. 6.** Analysis of the teaching strategy optimization effect of the vocal music teaching optimization model based on six datasets.

#### Analysis of the resource generation effect

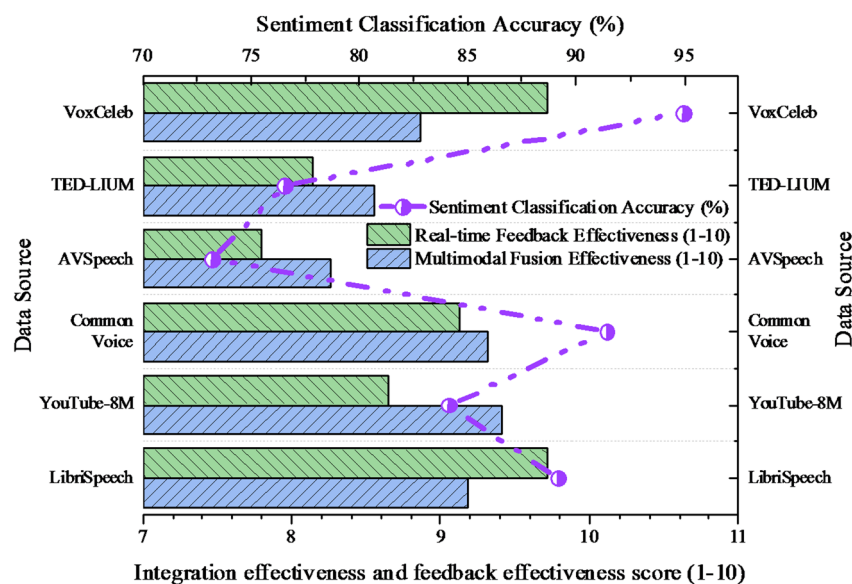
The analysis results of the resource generation effect of the vocal music teaching optimization model based on six datasets are indicated in Fig. 7:

Figure 7 underscores that the Common Voice dataset demonstrates significant advantages in Quality Score at 97.91% and User Satisfaction at 95.20%, indicating its superior quality of generated resources and user satisfaction. The TED-LIUM shows strong performance in Engagement Rate and Improvement Rate at 97.60% and 91.54%, indicating its potential to enhance student engagement and learning improvement. The VoxCeleb exhibits competitive scores in Realism Score at 93.14% and Variety Score at 90.72%, showcasing its strength in realism and diversity of generated resources. The LibriSpeech displays balanced performance across all indicators, including Resource Utilization at 89.25% and Improvement Rate at 88.15%, reflecting its stability and reliability in resource utilization efficiency and teaching effectiveness.





**Fig. 7.** Analysis of the resource generation effect of the vocal music teaching optimization model using six datasets.



**Fig. 8.** The comprehensive evaluation effect results of the vocal music teaching optimization model using six datasets.

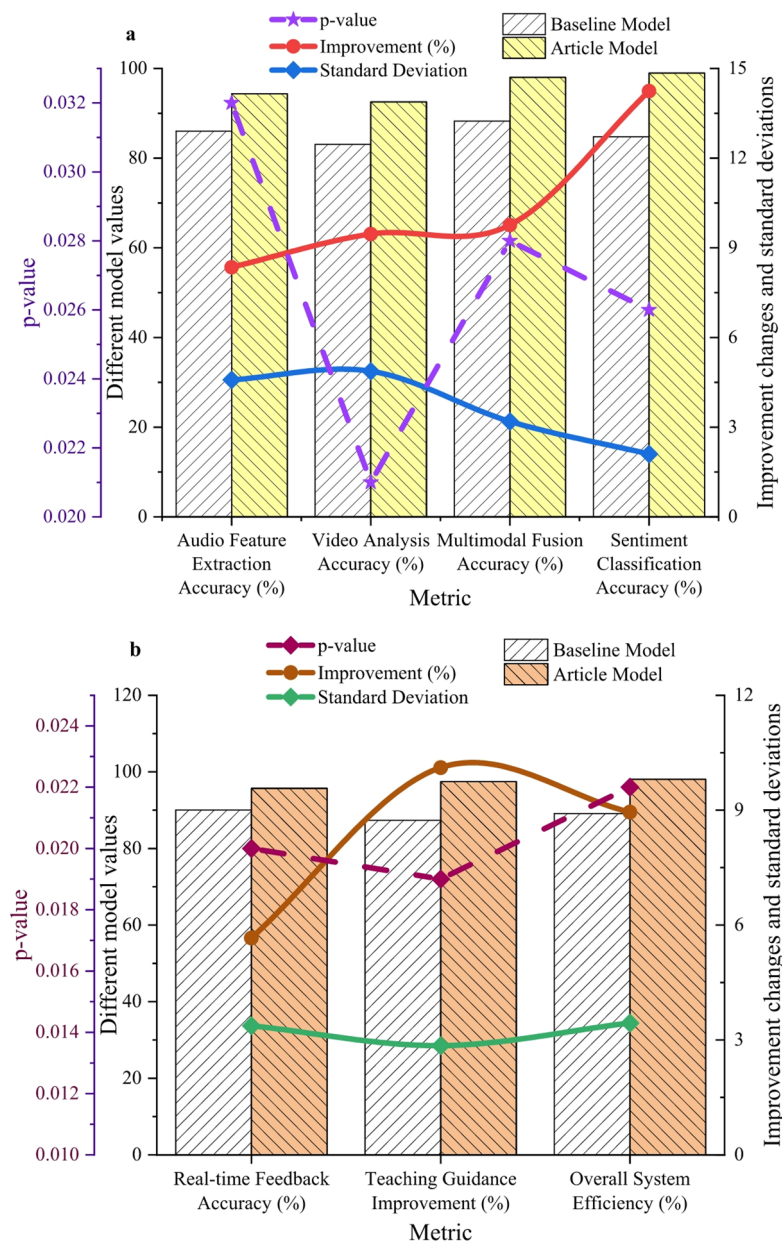
#### Comprehensive evaluation effect analysis

Figure 8 presents the comprehensive evaluation effect results of the vocal music teaching optimization model based on six datasets:

Figure 8 reveals that VoxCeleb performs best in real-time feedback effectiveness at 9.718 and sentiment classification accuracy at 94.98%, demonstrating its superiority in handling multimodal data. The Common Voice also shows impressive results in sentiment classification accuracy (91.42%) and real-time feedback effectiveness (9.131). LibriSpeech has a real-time feedback effectiveness of 9.715, indicating its efficiency in providing immediate teaching feedback. In contrast, AVSpeech performs weaker across all indicators, with a sentiment classification accuracy of only 73.19%. These results indicate that different data sources significantly influence the model's comprehensive evaluation effects. Particularly, these findings demonstrate the remarkable advantages of VoxCeleb and Common Voice in enhancing model performance and timely teaching feedback.

Figure 9 shows the comparative analysis of the comprehensive evaluation effects based on the proposed vocal music teaching optimization model and baseline models.

In Fig. 9, the vocal music teaching optimization model shows substantial differences compared with baseline models in multiple indicators. First, the audio feature extraction accuracy increases from 85.99% to 94.34%,



**Fig. 9.** Comparative analysis of the comprehensive evaluation effects of the proposed model and baseline models ((a): Comparative analysis of model performance indicators; (b): Comparative analysis of model teaching effect indicators).

with an improvement of 8.35%, and the p-value is 0.032, which is statistically significant. The video analysis accuracy rises from 83.09% to 92.55%, an increase of 9.46%, and the p-value is 0.021, which proves the improved model's effectiveness. The multimodal fusion accuracy rises from 88.26% to 98.02%, an improvement of 9.76%, and the p-value is 0.028, showing the effect of multimodal fusion technology. The sentiment classification accuracy increases from 84.75% to 98.99%, a rise of 14.24%, and the p-value is 0.026, indicating the BERT model's remarkable improvement. Regarding real-time feedback accuracy, it adds from 90.05% to 95.71%, an improvement of 5.66%, and the p-value is 0.020, showing a notable improvement. The teaching guidance improvement rate and overall system efficiency rise from 87.34% to 89.12% to 97.45% and 98.07% respectively, with increases of 10.11% and 8.95%, and the p-values are 0.019 and 0.022. The above results show that the optimized improved model has prominent advantages in multimodal data processing, SA, and real-time feedback, which enhances teaching effects and system efficiency.

#### Generated resources and multi-dimensional SA

To further verify the application value of the proposed model in vocal music teaching scenarios, this study conducts extended experiments from two aspects. (1) It compares the role of traditional resources and GAN-generated resources in teaching optimization. (2) It expands the dimension of SA, extending from single

| Resource type         | Recommendation accuracy (F1) | Strategy stability (%) | Fusion accuracy (%) | System efficiency (%) |
|-----------------------|------------------------------|------------------------|---------------------|-----------------------|
| Traditional resources | 0.81                         | 89.72                  | 92.36               | 90.58                 |
| Generated resources   | 0.88                         | 97.24                  | 98.02               | 98.07                 |

**Table 9.** The results of the comparative experiment between the generated and traditional resources.

| Actual category | Prediction: Happiness | Prediction: Sadness | Prediction: Passion | Prediction: Calmness |
|-----------------|-----------------------|---------------------|---------------------|----------------------|
| Happiness       | 95.12                 | 2.34                | 1.25                | 1.29                 |
| Sadness         | 3.11                  | 93.47               | 2.02                | 1.4                  |
| Passion         | 2.26                  | 1.85                | 94.08               | 1.81                 |
| Calmness        | 1.72                  | 2.43                | 1.89                | 93.96                |

**Table 10.** Confusion matrix of multi-dimensional emotion classification (%).

positive-negative sentiment classification to multi-category and intensity recognition, and conducting systematic evaluation combined with a confusion matrix.

In the resource value evaluation, the experiment builds a “traditional resource group” and a “generated resource group” based on six public datasets: LibriSpeech, YouTube-8 M, Common Voice, AVSpeech, TED-LIUM, and VoxCeleb. The traditional resource group only uses original samples, while the generated resource group introduces GAN-generated extended materials on this basis. Both groups of experiments were run under the same conditions. Comparisons are made through indicators such as recommendation accuracy (F1-score), strategy stability, fusion accuracy, and system efficiency. Table 9 shows the results:

In Table 9, all indicators improve after the introduction of generated resources. Among them, the recommendation accuracy increases by 0.07, and the strategy stability rises by 7.52% points. This indicates that GAN plays an important role in alleviating data sparsity and enriching teaching materials. Further observation shows that the improvement ranges of fusion accuracy and system efficiency are close to 8% points. This demonstrates that generated resources expand data coverage and improve the effect of cross-modal feature alignment, making the model more robust when processing complex inputs. Such improvements mean that in practical teaching applications, students can obtain more continuous and diverse learning experiences. At the same time, learning bottlenecks caused by insufficient resources are reduced.

In the SA extension experiment, emotional categories are divided into four types: happiness, sadness, passion, and calmness. Multimodal features of audio, video, and text are used for classification and intensity recognition. The results are plotted in Table 10:

Table 10 reveals that the ML/SA model achieves high accuracy in recognizing all types of emotions, and performs particularly well in the “happiness” and “sadness” categories. Although there is a certain degree of confusion in the “passion” category, the overall accuracy still reaches 98.99%. This indicates that the model can effectively distinguish the main emotional types in vocal music singing. It further shows that there is a close connection between emotion recognition results and artistic expression evaluation, providing solid data support for emotional feedback in the teaching process.

To sum up, the extension experiments show that generated resources improve the performance of recommendation and teaching optimization. Generated resources also enable the model to capture more fine-grained emotional categories and intensities in multimodal emotion recognition, closely related to artistic expression. This result further proves the proposed model’s teaching value and art education adaptability in vocal music teaching, offering a verifiable implementation path for interdisciplinary integration under the STEAM concept.

Discussion

Among existing cutting-edge methods, the MusicARLtrans Net model proposed by Chang et al. in 2024 mainly realized real-time feedback through speech-to-text and RL. The KNIGHT framework proposed by Sharif and Uckelmann in 2024 emphasized multimodal data and privacy protection. The DPN-GAN proposed by Ahmad et al. in 2025 focused on improving the fidelity of audio generation. Although these studies have made breakthroughs in interaction, analysis, or resource generation, they generally target a single task or local optimization. In contrast, the proposed model is not a simple juxtaposition of NCF, DQN, and GAN. Instead, guided by the inherent STEAM concept, it deeply integrates science, technology, engineering, arts, and mathematics. From the perspective of science and learning, the model defines learning status through quantitative indicators of intonation, rhythm, timbre, and emotion, converting the traditional experience-dependent evaluation into observable data. Regarding mathematical principles, the unified embedding space enables the comparison and calculation of features from different modalities, thus transcribing subjective artistic aesthetics into an optimizable objective function. At the technical and engineering level, the model forms a closed loop of generation, recommendation, and strategy optimization: GAN generates targeted vocal music segments; NCF realizes personalized ranking between real and generated resources; DQN dynamically balances accuracy, stability, and participation through reward shaping. At the artistic and educational level, the system trains skills and attaches importance to emotional expression and style presentation; this ensures that learners

gain a complete music experience in practice. This interdisciplinary design allows students to understand sound in scientific analysis, receive interventions in engineering systems, experience reasoning in mathematical modeling, and realize expression in artistic creation, reflecting a true STEAM learning path. The innovation of this study lies in connecting recommendations and strategies through measurable states and controllable generation. This allows artistic evaluation to be based on verifiable computational representations rather than a simple stack of single technologies.

Combined with the study's analysis results, it can be found that the proposed system performs well at the numerical level and has clear significance in teaching practice. The strategy stability is improved to 97.24%, which means the system can maintain sensitivity and consistency to students' states during long-term operation. As a result, teachers can reduce repeated trials and frequent adjustments and focus on guiding artistic expression. The cumulative reward reaches 90.23, indicating that students' performance in multiple dimensions, such as intonation, rhythm, timbre, and emotion, has achieved continuous improvement. This provides learners with a more traceable and verifiable growth path. In other words, the experimental indicators do not reflect abstract computational advantages but correspond to continuous progress and reduced intervention burden in real teaching. This highlights the system's value in providing two-way support for teachers and students.

From the perspective of real application scenarios, the model may face some challenges during actual deployment. First, complex multimodal computing imposes high requirements on computing power. In environments with limited resources, this may cause delays and affect the experience of real-time feedback. Second, the collection and annotation of cross-modal data require high costs, which may restrict the model's promotion in different schools or educational stages. Third, the model has great differences in adaptability between K-12 education and professional music colleges; the former requires more lightweight and universal approaches, while the latter emphasizes precision and personalization. Future research can enhance scalability through model compression, delay optimization, and cross-scenario transfer. This study also suggests that generative AI should be combined with domain prior knowledge and educational rules to avoid instability caused by completely free generation. By embedding constraints such as rhythm, vocal range, or textbook boundaries, the model can achieve output that is reliable and creative. Thus, it can truly balance innovation and practicality in educational environments.

## Conclusion

### Research contribution

The related research in Sect. “[Research background and motivations](#)” indicates that existing STEAM education models (e.g., Zhang, 2025; Sharif and Uckelmann, 2024) mainly focus on interdisciplinary knowledge integration and creative thinking cultivation. They promote the integration of art and science at the level of teaching concepts, but are still insufficient in terms of intelligent optimization and dynamic feedback mechanisms at the algorithm level. In addition, studies in the field of recommendation systems (e.g., Prabhakar and Lee, 2023; Chango et al., 2022; Oroojlooyjadid et al., 2022) have made remarkable progress in personalized resource matching and learning behavior modeling. However, they generally ignore the complexity of emotional expression, multimodal interaction, and aesthetic experience in art education. Most existing achievements concentrate on task performance optimization, while lacking the systematic integration of human-computer collaboration and emotional feedback in educational scenarios.

In contrast, this study proposes a vocal music teaching optimization model based on ML and SA by integrating the STEAM concept into the intelligent recommendation system. Experimental results show that this model is markedly superior to existing methods in recommendation accuracy, teaching strategy optimization, resource generation, and real-time feedback. Meanwhile, this model provides a reference solution for the personalization and intellectualization of vocal music education. The proposed intelligent recommendation-based vocal music teaching optimization framework, driven by the STEAM concept, deeply couples educational concepts, affective computing, and ML. It realizes the dynamic linkage of teaching strategy optimization, resource generation, and emotional feedback. Moreover, it makes up for the disconnection between “educational concepts-algorithm models-teaching practice” in existing studies and reflects innovative value at the methodological and application levels.

In practical teaching application scenarios, the deployment and promotion of this system need to fully consider the differentiated conditions of educational platform ecology and hardware environment. In real vocal music teaching, different colleges and online education platforms have diverse infrastructure and data interface standards. When connecting with existing Learning Management Systems (LMS) or Content Management Systems (CMS), the system needs to achieve multi-protocol compatibility and interface adaptation to ensure the seamless integration of recommendation and feedback modules. In addition, in teaching environments with limited resources or low network bandwidth, the computational load of real-time multimodal feedback may cause delays, thereby affecting the consistency and interactivity of the teaching experience. Therefore, during actual deployment, the model can combine edge computing or cloud collaboration strategies and reduce computational delay through local caching and distributed reasoning. At the same time, regarding the usage differences between teachers and students, the system should provide a visual feedback dashboard and a simplified mobile push mechanism to support flexible interaction and real-time feedback in different teaching scenarios, thereby achieving in-depth integration of technological innovation and teaching practice.

### Future works and research limitations

It should be noted that the results of this study should be regarded as a preliminary exploration. On the one hand, although the experimental data support the model's effectiveness, further verification is still needed in larger-scale and more diverse educational scenarios. On the other hand, the current model still relies on complex deep structures for real-time alignment of multimodal features and cross-modal semantic consistency, resulting



in high computational overhead. It is necessary to consider further improving the model's computational efficiency and response speed while maintaining its performance. Thus, it has greater feasibility and stability in subsequent deployment in large-scale real-time teaching scenarios. In other words, this study mainly offers a feasible technical framework and research direction, rather than a fully mature conclusion.

Future research can optimize multimodal data fusion technology to improve the model's robustness and adaptability when dealing with extreme data and long-tail user behaviors. At the same time, future research can combine lightweight multimodal architecture with adaptive constraint mechanisms to enhance the model's generalization ability and interpretability. It can also explore the application of explainable artificial intelligence methods in teaching feedback and human-computer collaboration, to realize a more efficient, transparent, and scalable intelligent vocal music teaching system.

## Data availability

The datasets used and/or analysed during the current study available from the corresponding author Qianping Guo on reasonable request via e-mail 15340524767@163.com.

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## Author contributions

Fanbo Su: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation Qianping Guo: writing—review and editing, visualization, supervision, project administration, funding acquisition.

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## Declarations

## Competing interests

The authors declare no competing interests.

## Ethics statement

The studies involving human participants were reviewed and approved by College of Music, Chongqing Normal University Ethics Committee (Approval Number: 2022.25641000). The participants provided their written informed consent to participate in this study. All methods were performed in accordance with relevant guidelines and regulations.

## Additional information

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